

Installing petalinux 2024.2 and configuring/building linux for our ZedBoard

- We used Vitis Unified 2024.2 to generate our platform component 'platform_zed_board_linux', which contains the 'hw' output folder and zed.xsa file (reference 1_Steps.pdf)
- We have Ubuntu 22.04 LTS running via WSL2 on our Windows 11 PC
- This document is for:
 - Installing the petalinux tools in our Ubuntu VM,
 - Using the petalinux tools to configure and build the linux kernel for our ZedBoard.

Petalinux Download: <https://www.xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/embedded-design-tools/2024-2.html>

The screenshot displays the Xilinx website page for downloading Petalinux 2024.2. The browser address bar shows the URL: [xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/embedded-design-tools/2024-2.html](https://www.xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/embedded-design-tools/2024-2.html). On the left, a 'Version' dropdown menu is open, showing options: 2025.1, 2024.2 (selected), 2024.1, and Archive. The main content area features a red 'Important Information' section stating that starting with the 2024.1 release, some Petalinux BSPs are entering maintenance mode. Below this, it mentions that pre-built images, source code, and configurations are provided for demonstration purposes only. A section titled 'PetaLinux Tools - Installer - 2024.2 Full Product Installation' contains an 'Important Information' box explaining that the installer checks for host machine requirements and that all BSPs require the PetaLinux Tools to be installed first. A download link for 'PetaLinux installer (TAR/GZIP - 2.94 GB)' is provided, along with the MD5 SUM Value: 1c88abb1ba23370692c0aa9d27c0e74. On the right side of this section, a table lists metadata: Download Type (Full Product Installation), Last Updated (Nov 18, 2024), Answers (Release Notes and Known Issues), and Documentation (PetaLinux Tools Documentation, Installer Information, What's new in Embedded Software).

PetaLinux Tools - Installer - 2024.2 Full Product Installation	
Download Type	Full Product Installation
Last Updated	Nov 18, 2024
Answers	Release Notes and Known Issues
Documentation	PetaLinux Tools Documentation Installer Information What's new in Embedded Software

Downloaded file: petalinux-v2024.2-11062026-installer.run

Copied installer to Ubuntu

In Ubuntu command line:

- Do these steps before installing petalinux:
 - 73296 - PetaLinux: How to install the required packages for the PetaLinux Build Host:
 - Download 'plnx-env-setup.sh' from 'https://adaptivesupport.amd.com/s/article/73296?language=en_US'
 - Copy script to Ubuntu
 - '[chmod +x plnx-env-setup.sh](#)'
 - '[sudo ./plnx-env-setup.sh](#)'
 - Disable 'dash':
 - Up to Ubuntu 22.04 included (and Debian 11 included):
 - '[sudo dpkg-reconfigure dash](#)' (answer 'No' to transition to bash shell)
 - **Or**, for above version Ubuntu 22.04, try:
 - [sudo ln -s bash /bin/sh.bash](#)
 - [sudo mv /bin/sh.bash /bin/sh](#)
 - And if/when you want to restore back to dash:
 - [sudo ln -s dash /bin/sh.dash](#)
 - [sudo mv /bin/sh.dash /bin/sh](#)
 - Prevention of exception due to 'warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8):'
 - [sudo apt install locales](#)
 - [sudo locale-gen en_US.UTF-8](#)
 - [sudo dpkg-reconfigure locales](#) (make sure 'en_US.UTF-8' is starred)
 - Prevention of exception due to 'package require xsbd FAILED':
 - [sudo apt update](#)
 - [wget http://security.ubuntu.com/ubuntu/pool/universe/n/ncurses/libtinfo5_6.3-2ubuntu0.1_amd64.deb](#)
 - [sudo apt install ./libtinfo5_6.3-2ubuntu0.1_amd64.deb](#)
- Installing petalinux:

- **'chmod +x petalinux-v2024.2-11062026-installer.run'**
- **'mkdir -p /home/wlaba/playZed/petalinux-tools'**
- **'./petalinux-v2024.2-11062026-installer.run -d /home/wlaba/playZed/petalinux-tools'**

Installer output:

./petalinux-v2024.2-11062026-installer.run: line 3: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8): No such file or directory
PetaLinux CMD tools installer version 2024.2

```
=====
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
[WARNING] This is not a supported OS
[INFO] Checking free disk space
[INFO] Checking installed tools
[INFO] Checking installed development libraries
[INFO] Checking network and other services
[WARNING] No tftp server found - please refer to "UG1144 PetaLinux Tools Documentation Reference Guide" for its impact and solution
```

LICENSE AGREEMENTS

PetaLinux SDK contains software from a number of sources. Please review the following licenses and indicate your acceptance of each to continue.

You do not have to accept the licenses, however if you do not then you may not use PetaLinux SDK.

Use PgUp/PgDn to navigate the license viewer, and press 'q' to close

Press Enter to display the license agreements

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/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
Do you accept Xilinx End User License Agreement? [y/N] > y
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
Do you accept Third Party End User License Agreement? [y/N] > y
[INFO] Installing PetaLinux SDK...done
[INFO] Setting it up...
```

[INFO] Extracting xsct tarball...

:
:

[INFO] PetaLinux SDK has been successfully set up and is ready to be used.

cd to your installation folder (the folder given after the -d option)

source ./settings.sh

The PetaLinux source code and images provided/generated are for demonstration purposes only.

Please refer to <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/2741928025/Moving+from+PetaLinux+to+Production+Deployment> for more details

PetaLinux environment set to '/home/wlaba/playZed/petalinux-tools'

WARNING: /bin/sh is not bash!

bash is PetaLinux recommended shell. Please set your default shell to bash.

[WARNING] This is not a supported OS

[INFO] Checking free disk space

[INFO] Checking installed tools

[INFO] Checking installed development libraries

[INFO] Checking network and other services

[WARNING] No tftp server found - please refer to "UG1144 2024.2 PetaLinux Tools Documentation Reference Guide" for its impact and solution

Create our first project:

- **cd /home/wlaba/playZed/petalinux-tools** (change directory to our install folder)
- **source ./settings.sh**
- **petalinux-create --type project --template zynq --name PetalinuxDemo1**

Tool output:

[NOTE] Argument: "--type project" has been deprecated. It is recommended to start using new python command line Argument.

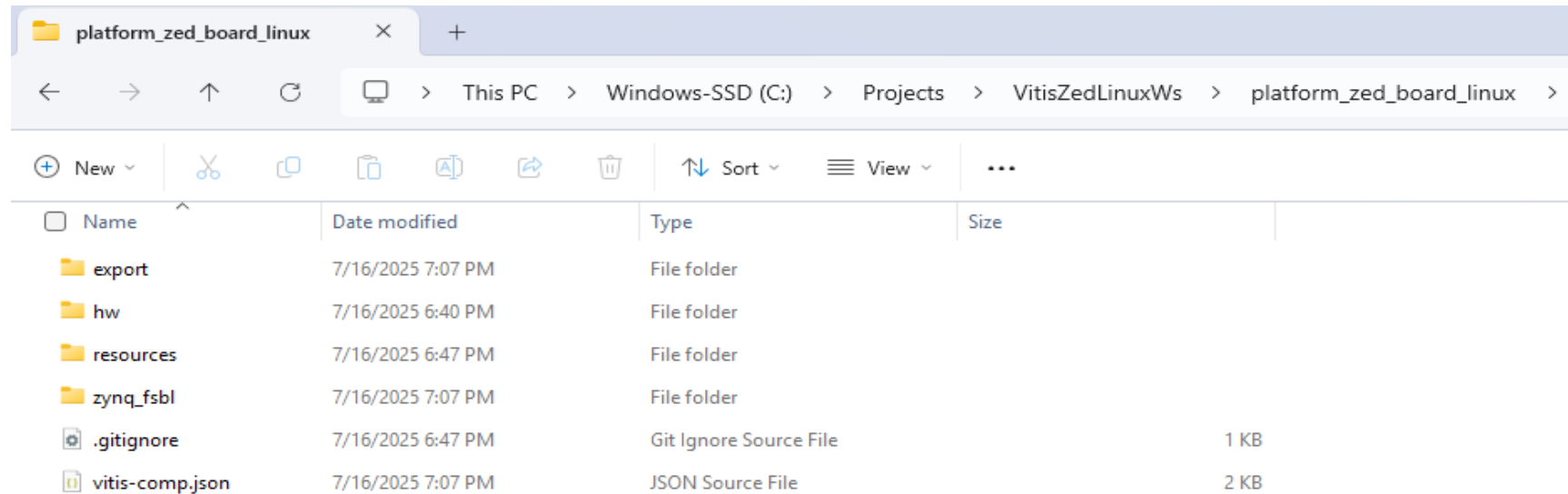
[NOTE] Use: petalinux-create project [OPTIONS]

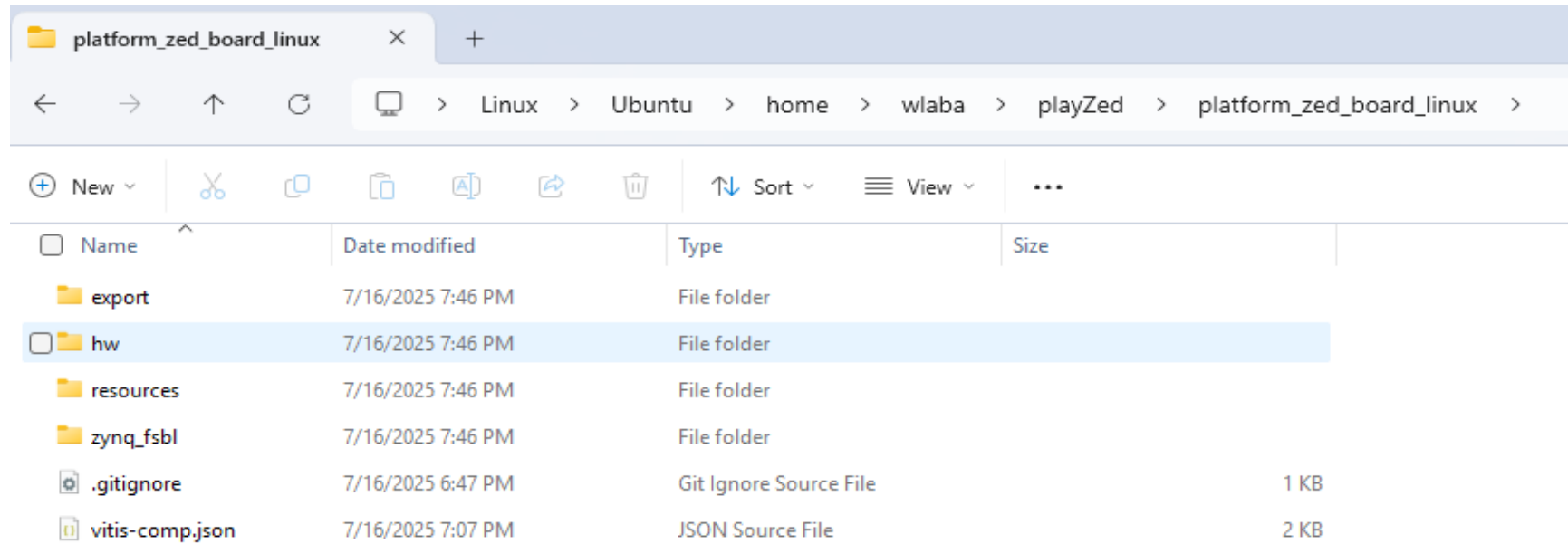
[INFO] Create project: PetalinuxDemo1

[INFO] New project successfully created in /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1

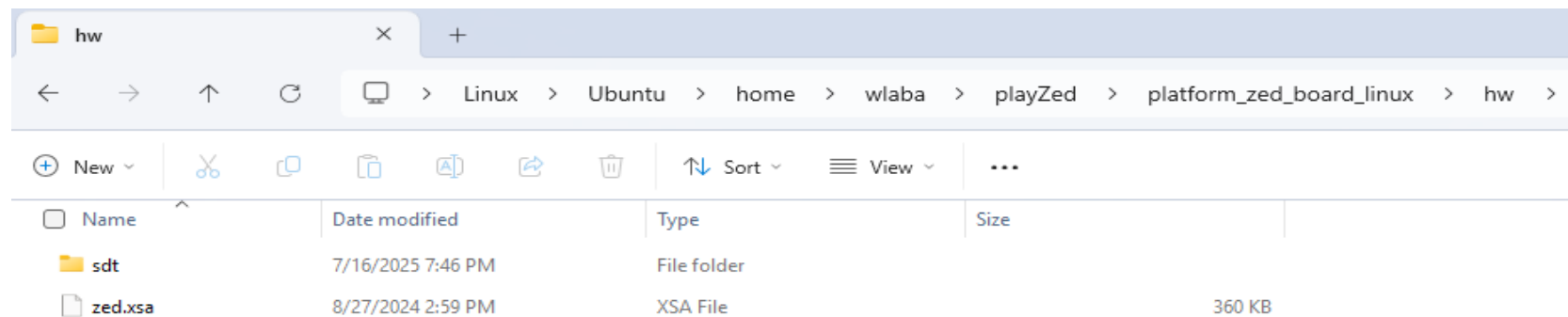
Create our first hardware description output:

- Using 'platform_zed_board_linux' folder generated by Vitis in '1_Steps.pdf'
- Copied Vitis output folder 'platform_zed_board_linux' from windows to Ubuntu folder:





- The 'zed.xsa' file and 'sdt' folder were located under the 'hw' sub-folder:



- `cd /home/wlaba/playZed/petalinux-tools/PetalinuxDemo1` (change directory to our project folder)
- `petalinux-config --get-hw-description=/home/wlaba/playZed/platform_zed_board_linux/hw/`

Tool output:

INFO] Getting hardware description

[INFO] Renaming **zed.xsa** to system.xsa

:

< Insert Successful Configuration artifacts HERE! >

Perform our first petalinux build:

- **'petalinux build'** (continuing from within our project folder)

Tool output:

:

< Insert Successful Build artifacts HERE! >

Summary:

- Looks like petalinux 2024.2 was installed on Ubuntu successfully
- Looks like petalinux project was created successfully
- Petalinux configure succeed and we see the menu config screen!
- Results of Petalinux build - tbd??